Supervised Learning algorithms

Regression(hồi quy)

Linear Regression
Polynomial Regression

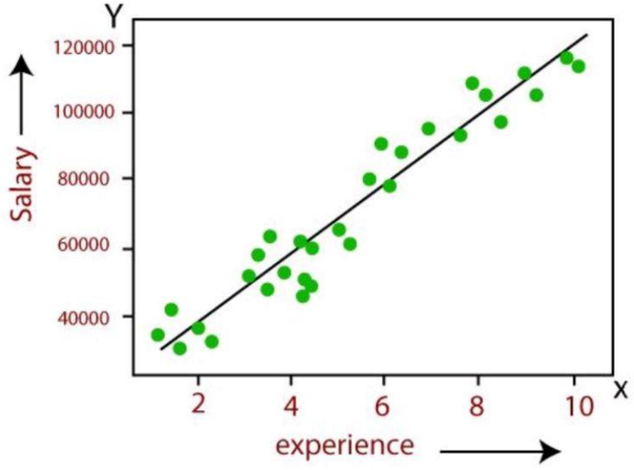
*Could also be used here

Classification(Phân loại)

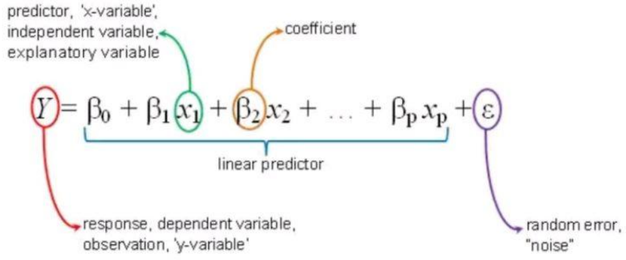
Logistic Regression
Support Vector Machine*

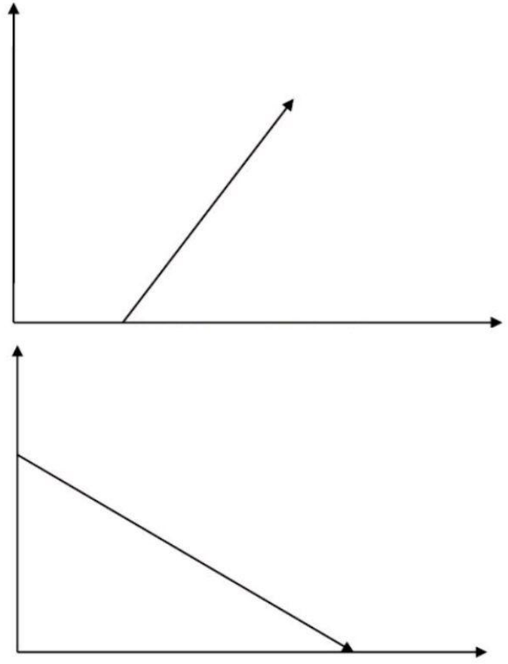
K-nearest neighbours*
Naive Bayes
Decision Tree*
Random Forest*

Linear Regression

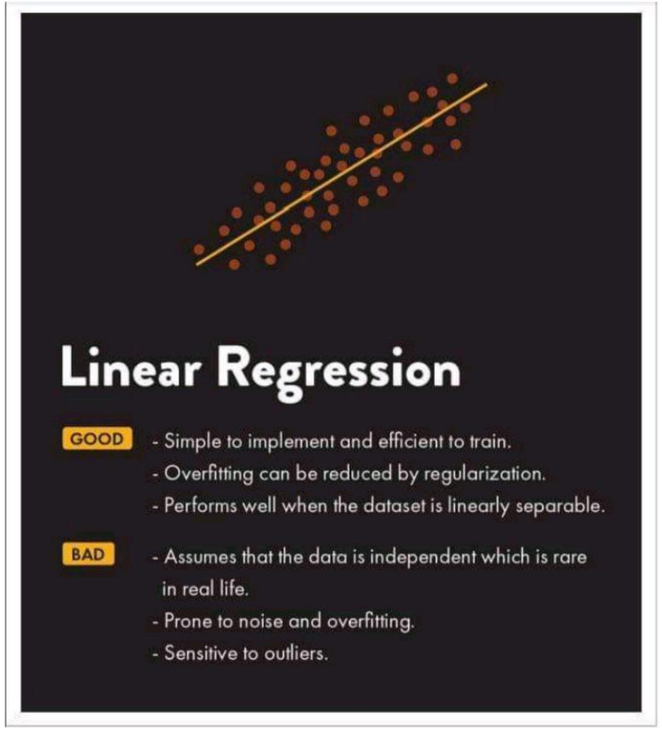


- 1 trong số các thuật toán cơ bản và đơn giản nhất trong Supervised Learning
- Simple linear regression: 1 input feature
- Multiple linear regression: 2 input features trở lên
- Phù hợp với các linear dataset

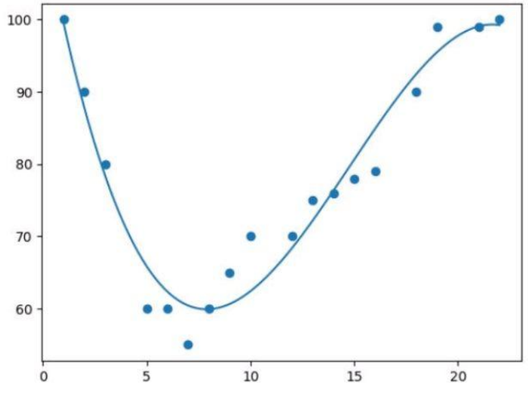


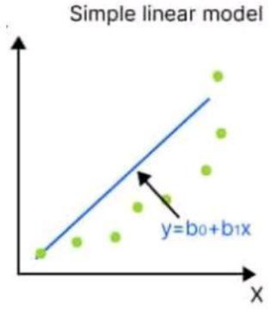


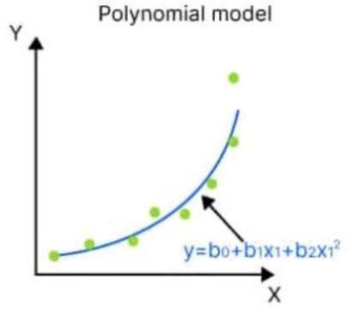
Negative Linear Relationship



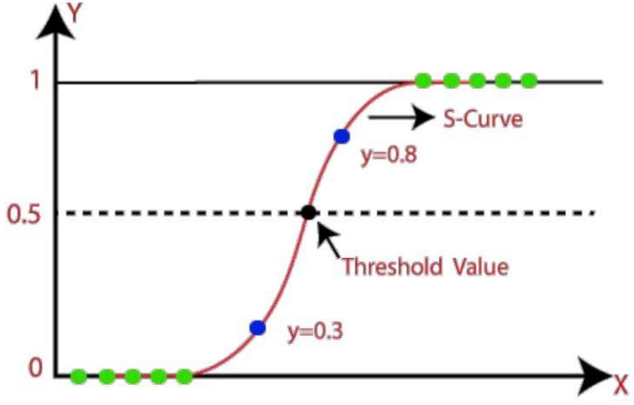
Polynomial Regression

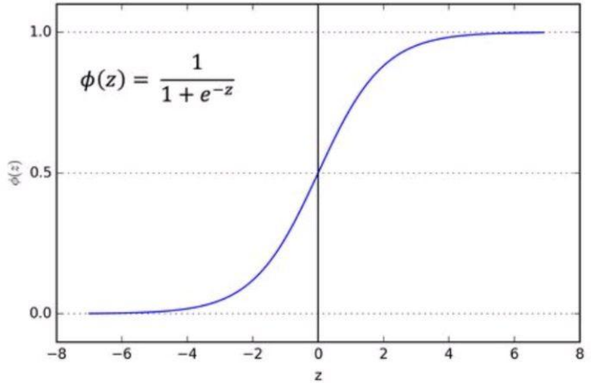


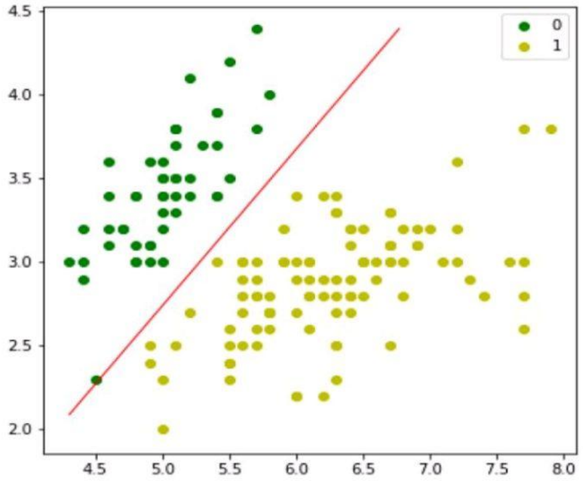


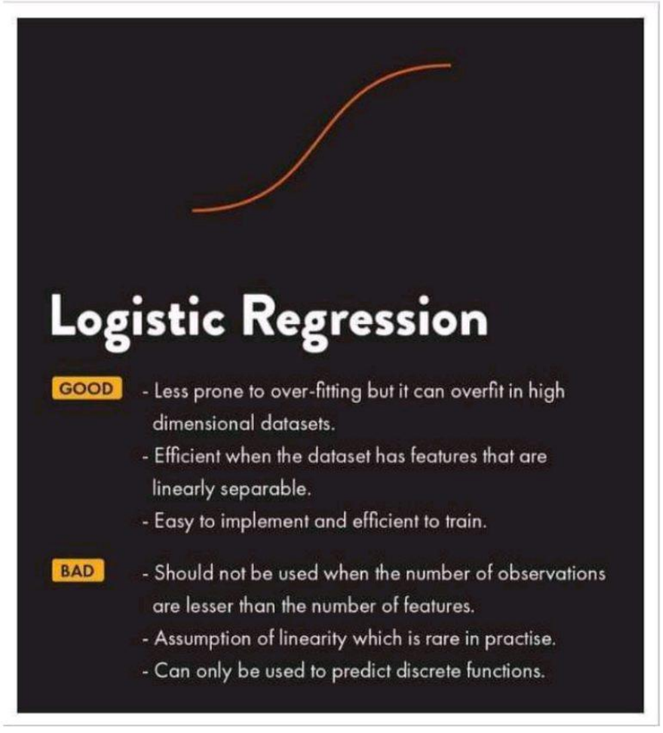


Logistic Regression






4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
0
1


Logistic Regression
GOOD
- Less prone to over-fitting but it can overfit in high dimensional datasets.
- Efficient when the dataset has features that are linearly separable.
- Easy to implement and efficient to train.
BAD
- Should not be used when the number of observations are lesser than the number of features.
- Assumption of linearity which is rare in practise.
- Can only be used to predict discrete functions.